



UTM

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SECTION 01

RETAIL SALES AND INVENTORY MANAGEMENT ANALYSIS

PHASE 1: INTRODUCTION

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1. Problem Background

Business Intelligence (BI) plays an important role in helping organizations analyze large amounts of data and transform it into meaningful insights for better decision-making. In the retail industry, businesses generate data from various sources such as sales transactions, inventory records, and supplier information. However, these datasets are often stored separately, making it difficult for companies to gain a complete overview of their operations.

- **Business context**

The e-Commerce industry is known for its wide and diverse products and locations and a significant volume of transactions made. Olist, one of the leading Brazilian department markets that specialized in bridging the gap between small businesses to large businesses. However, there is a challenge in getting the synergy between seller locations, product categories and delivery logistics. The information on sales velocity, logistics cost and regional demand is currently scattered in isolated data sets, and it is challenging for stakeholders to get a holistic picture of the business ecosystem.

- **Importance of the problem**

For retail organizations, poor inventory management, especially stockouts and overstocking, can result in serious operational and financial issues. Stockouts occur when products are unavailable when customers want to purchase them, resulting in missed sales opportunities and lower customer satisfaction. Overstocking occurs when businesses keep excessive inventory that exceeds customer demand, resulting in operational waste and increase in storage costs. In addition, delays in supplier deliveries may negatively impact inventory planning and product availability. As a result, businesses may struggle to maintain between meeting customer demand and reducing inventory-related costs.

- **Expected impact**

The expected impact of this project is to create a clear map of the Olist's fragmented data to develop a better understanding. The financial loss from excess stock and missed sales, due to stock cycles, will be avoided by creating an integrated BI solution to ensure that the right products are available for the customer. Rather than trying to piece together data,

the stakeholders will be able to visualise the details of delivery and logistics cost on interactive dashboards to see where it is taking place and how it is eating into profits. This project will help the company go from guess to know by leveraging an automated Alteryx workflow to offer quick, accurate insights that can help enhance seller cooperation and general customer satisfaction.

2. Objectives

The objectives for the implementation of this project is as follows:

- To integrate disparate datasets (Sales, Products, and Suppliers) into a unified data warehouse using Alteryx Designer.
- To analyze sales performance, inventory trends, and supplier operations in order to identify patterns, inefficiencies, and potential business risks.
- To develop interactive data visualizations and dashboards that provide meaningful insights into retail operations and support data-driven decision-making.
- To enhance inventory planning and retail performance by reducing stock-related issues and improving overall business efficiency.

3. Datasets Description

To meet the requirement of using relevant dataset, the Brazilian E-Commerce Public Dataset by Olist is utilised as it provides multiple interconnected tables where the dataset is obtained from the open-source Kaggle. The description for each dataset used is given as below.

1. Brazilian E-Commerce Orders Dataset

- **Source:** [Kaggle - Olist E-Commerce \(Order\)](#)
- **Type of data**

This dataset is categorized as structured data because order records are stored in a well-organized tabular format.

- **Key variables/features**

Variable Name	Description
order_id	Unique identifier for each transaction.

customer_id	Key to link orders to specific customer locations.
order_status	Status of the order to track stockouts.
order_purchase_timestamp	The date and time for the purchase was made for trending analysis.
order_delivered_customer_date	The actual arrival date to measure logistics performance.

2. Brazilian E-Commerce Products Dataset

- **Source:** [Kaggle - Olist E-Commerce \(Products\)](#)

- **Type of data**

This dataset is categorized as structured data because product records are stored in a well-organized tabular format.

- **Key variables/features**

Variable Name	Description
product_id	Unique identifier for each product.
product_category_name	Name of the product category for sales analysis.
product_weight_g	Product weight used to calculate shipping and logistics impact.
product_length_cm	Length of the product for inventory and storage planning.

3. Brazilian E-Commerce Sellers Dataset

- **Source:** [Kaggle - Olist E-Commerce \(Sellers\)](#)

- **Type of data**

This dataset is categorized as structured data because seller details are stored in a well-organized tabular format.

- **Key variables/features**

Variable Name	Description
seller_id	Unique identifier for the seller.
seller_zip_code_prefix	Location code to identify the geographic sales demand.
seller_city	City where the seller is located.
seller_state	State where the seller is located for regional comparison.

4. Brazilian E-Commerce Order Item Dataset

- **Source:** [Kaggle - Olist E-Commerce \(Order Items\)](#)

- **Type of data**

This dataset is categorized as structured data because order item records are stored in a well-organized tabular format.

- **Key variables/features**

Variable Name	Description
order_id	Unique identifier for each order.
product_id	Unique identifier for each product.
seller_id	Unique identifier for the seller.
price	The cost of items that used to calculate total sales and revenue.
freight_value	The shipping cost used to analyze logistics expenses and profit margins.

4. Proposed BI Approach

The aim of this project is to build a Business Intelligence (BI) system to transform the raw data into clear information. Complex sales and shipping records are made simple using Alteryx Designer to clean and integrate information and Microsoft Power BI to create interactive dashboards. The main goal is to automate data preparation to allow managers to easily spot delays in delivery and any inventory problems in one single integrity interface.

Technology Used

This project will utilize Alteryx Designer for Extract, Transform, and Load (ETL) process. For the development of the analytical dashboard for business insight, it will be developed by using Microsoft Power BI to create an interactive dashboard without spending any money.

Data Integration, Cleaning & Transformation

To ensure the data integrity is maintained while preparing the datasets for analysis that will be designed in Alteryx Designer, the following method is applied:

a. Data Integration

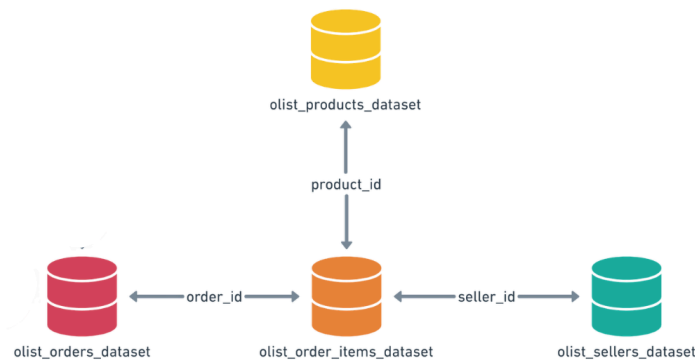


Figure 1: Data Schema (Source: [Kaggle - Brazilian E-Commerce Public Dataset by Olist](#))

A multiple stage Join Tool process will be used where the first join is Orders and Order Items will be joined via order_id. Then, the result will be joined with Products via product_id and Sellers via seller_id. This creates a flat, "master" analytical table.

b. Data Cleaning

In order to maintain the consistency and integrity of the data, the Filter Tool will be utilized by excluding the null values in critical fields like price or category names of datasets. This also will ensure the BI analysis reflects actual revenue.

c. Data Transformation

The Formula Tool will be utilized in creating a new feature, such as "Delivery Duration" by subtracting `order_purchase_timestamp` from `order_delivered_customer_date` to calculate the actual delivery lead time. The Summarize Tool will be employed to aggregate data by `seller_state` and `order_purchase_timestamp` to prepare for trend and comparative analysis.

Planned Analysis and Visual Forms

The top visuals for analysis in Microsoft Power BI that are best suited to address the issues of stockout, overstocking, and logistics expenses are:

1. Line and Cluster Column Chart

Used to perform the trend analysis over time. This will produce a graph of Total Revenue (Columns) as Average Delivery Days (Line). Drilling down from an annual level to the monthly level, users can determine whether there is a correlation between monthly sales increases and any logistics issues, such as delays or potential stockouts. Figure 2 showing the example of Line and Cluster Column Chart.

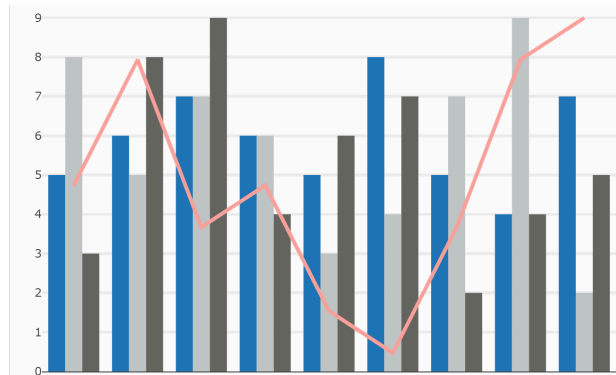


Figure 2: Example of Line and Cluster Column Chart

(<https://share.google/hBshPqBouADW9JTVe>)

2. Decomposition Tree

AI-driven visual tool for stakeholders to do a Root Cause Analysis. For instance, the revenue is low or the logistics cost is high, a manager can break down the data by State and the City and Product Category to identify where the business synergy is failing. Figure 3 showing the example of the Decomposition Tree in MS Power BI.

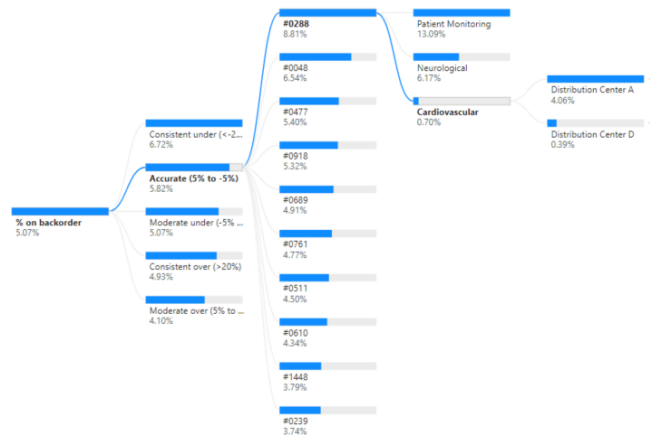


Figure 3: Example of Decomposition Tree (<https://share.google/rnHVpanOvliPjmxD0>)

3. Map Visualizations

Used to analyze seller performance geographically. The seller locations will be denoted by a bubble and the size of the bubble will represent the Total Freight Value. This enables the organization to clearly identify high shipping costs that consume high profit. Figure 4 shows the example of Map Visualization or Bubble Map that has different size of bubbles.

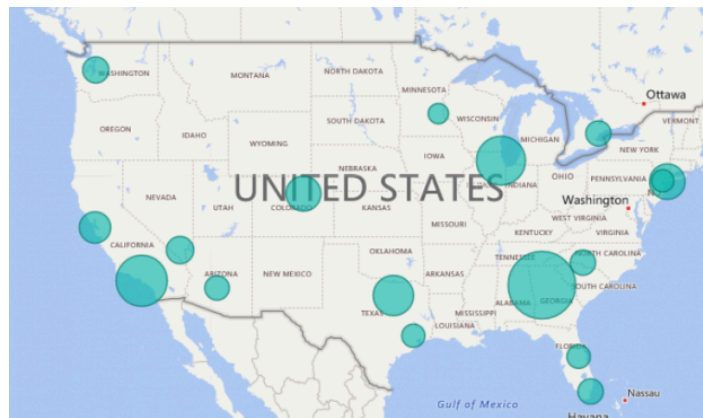


Figure 4: Example of Map Visualization (<https://share.google/jSAL40gZPQvbsNOrD>)

4. Conditional Formatting

Used to create a Dynamic Inventory Health table. Users can see at a glance whether there are risks of stock out or overstock situations for their products based on a colour gradient to different product groups. Figure 5 showing the example of the Conditional Formatting.

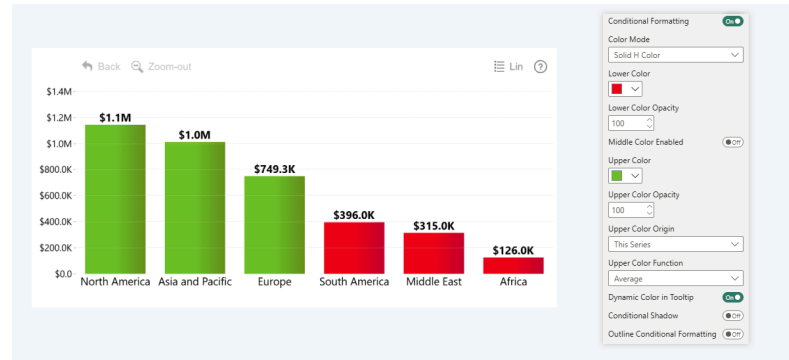


Figure 5: Example of Conditional Formatting (<https://share.google/3cfasY4FdY3cuznrE>)

5. KPI Cards

A high-level overview of business health. It will be the real-time metrics which will be shown on the cards for example Total Orders, Average Item Price, and Total Freight Cost. These cards are the main decision support document that enables executives to track what they should see the BI solution do as shown in example in Figure 6.

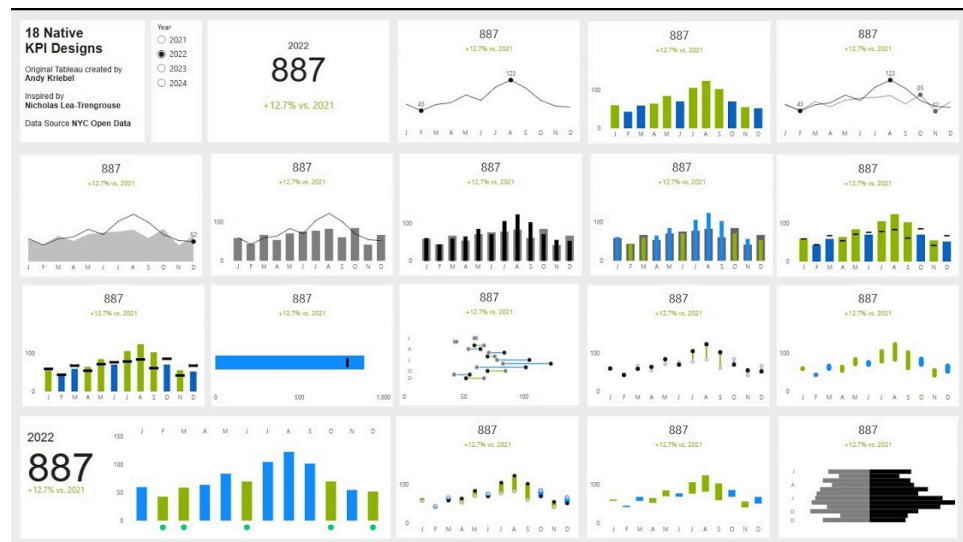


Figure 6: Example of KPI Cards (<https://share.google/GaTWJJd7WU7cX8MOY>)